

PENGFEI ZHOU

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EDUCATION

Doctor of Philosophy School of Computing National Univeristy of Singapore , Singapore	Advisor: Prof. Yang You Graduating: Dec. 2028 GPA: 5.0/5.0, NUS Research Scholarship
Master of Engineering Institute of Computing Technology Chinese Academy of Sciences , Beijing, China	Advisor: Prof. Weiqing Min and Prof. Shuqiang Jiang Graduated: Jul. 2024 GPA: 3.62/4.0
Bachelor of Engineering College of Computer Science Zhejiang University of Technology , Hangzhou, China	Advisor: Prof. Cong Bai Graduated: Jun. 2021 Grade: 86.0/100 (<i>Ranking: 2/46</i>)

MAJOR AWARDS & HONORS

Top-tier

- CSIG Outstanding Graduate (Master Thesis Incentive Award, *Top 10% of all applicants*) Nov. 2025
- Outstanding Graduate of Beijing (*First Class Honors, 10K/200K+ in Beijing*) May 2024
- Outstanding Graduate of Zhejiang Province (*First Class Honors, 10K/300K+ in Zhejiang*) May 2021
- **National Scholarship** (*3/300+ in VIPL Group and 25/1K+ in ICT, CAS*) Dec. 2023
- **National Scholarship** (*5/1K+ in College of Computer Science and 50/4K+ in ZJUT*) Dec. 2020

International

- Silver Award in Zhejiang International "Internet+" Undergraduate Entrepreneurship Competition Sep. 2021
- Second Place in International Information Retrieval Competition of LMRT in ImageCLEF Lifelog 2019 Sep. 2019

National

- E-Funds Fintech Fellowship by ICT, Chinese Academy of Sciences (*Top Honors, 10/300+ in ICT*) Jan. 2023
- "Jianxing" Honors Student by Zhejiang University of Technology (*Top Honors, 86/4K+ in ZJUT*) Jun. 2021
- First Prize in "HUAWEI Cup" National Undergraduate IoT Design Contest Aug. 2020
- Second Prize in National Service Outsourcing Competition Aug. 2020

PUBLICATIONS

Representative

- **Pengfei Zhou**, Jie Xia, Xiaopeng Peng, Wangbo Zhao, Zilong Ye, Zekai Li, Suorong Yang, Jiadong Pan, Yuanxiang Chen, Ziqiao Wang, Kai Wang, Qian Zheng, Xiaojun Chang, Gang Pan, Shurong Dong, Kaipeng Zhang, Yang You. 2025. *Neural-Driven Image Editing*. Advances in Neural Information Processing Systems (*24.52% Acceptance Rate, CORE A* Top-tier Conference*)
- **Pengfei Zhou**, Xiaopeng Peng, Jiajun Song, Chuanhao Li, Zhaopan Xu, Yue Yang, Ziyao Guo, Xiaojun Chang, Yu Qiao, Wenqi Shao and Kaipeng Zhang. 2025. *OpenING: A Comprehensive Benchmark for Judging Open-ended Interleaved Image-Text Generation*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**Oral**, *0.7% in 13K Submissions, CORE A* Top-tier Conference*)
- **Pengfei Zhou**, Fanrui Zhang, Xiaopeng Peng, Zhaopan Xu, Jiaxin Ai, Yansheng Qiu, Chuanhao Li, Zhen Li, Ming Li, Yukang Feng, Jianwen Sun, Haoquan Zhang, Zizhen Li, Xiaofeng Mao, Wangbo Zhao, Kai Wang, Xiaojun Chang, Wenqi Shao, Yang You, Kaipeng Zhang. 2025. *MDK12-Bench: A Multi-Discipline Benchmark for Evaluating Reasoning in Multimodal Large Language Models*. Proceedings of the AAAI Conference on Artificial Intelligence (*CORE A* Top-tier Conference*)
- **Pengfei Zhou**, Weiqing Min, Chaoran Fu, Ying Jin, Mingyu Huang, Xiangyang Li, Shuhuan Mei and Shuqiang Jiang. 2025. *FoodSky: A Food-oriented Large Language Model that Can Pass the Chef and Dietetic Examination*. Patterns (*Cell Press, JCR-Q1 Top-tier Journal*)

- **Pengfei Zhou**, Weiqing Min, Jiajun Song, Yang Zhang, and Shuqiang Jiang. 2024. *Knowledge-Enhanced Features Synthesizer for Real-World Zero-Shot Food Detection*. IEEE Transactions on Image Processing (JCR-Q1 Top-tier Journal)
- **Pengfei Zhou**, Weiqing Min, Yang Zhang, Jiajun Song, Ying Jin, and Shuqiang Jiang. 2023. *SeeDS: Semantic Separable Diffusion Synthesizer for Zero-shot Food Detection*. Proceedings of the ACM International Conference on Multimedia (29.3% Acceptance Rate, CORE A* Top-tier Conference)
- **Pengfei Zhou**, Kaining Ying, Zhenhua Wang, Dongyan Guo, and Cong Bai. 2022. *Self-Supervised Enhancement for Named Entity Disambiguation via Multimodal Graph Convolution*. IEEE Transactions on Neural Networks and Learning Systems (JCR-Q1 Top Journal)
- **Pengfei Zhou**, Cong Bai, Jie Xia, and Shengyong Chen. 2022. *CMRDF: A Real-Time Food Alerting System Based on Multimodal Data*. IEEE Internet of Things Journal (JCR-Q1 Top Journal)

Selected On Going & Under Review

- **Pengfei Zhou**, Hexin Wang, Yixing Ma, Wangbo Zhao, Kaipeng Zhang, and Yang You. *The Bitter Lesson of World Model*. To be submitted to Nature Machine Intelligence (Nature Sub-journal, JCR-Q1 Top-tier Journal)
- **Pengfei Zhou**, Yixing Ma, Zhenglin Wan, Wangbo Zhao, Kaipeng Zhang, and Yang You. *Routed On-Policy Distillation for Omni-Modal World Models*. To be submitted to ICLR 2027.
- **Pengfei Zhou**, Yixing Ma, Lama Moukheiber, Ziqiao Wang, Zhenglin Wan, Wangbo Zhao, Kaipeng Zhang, and Yang You. *Robust On-Policy Self-Distillation for Simulator World Models*. To be submitted to ICLR 2027.
- **Pengfei Zhou**, Zhiwei Tang, Yixing Ma, Jiasheng Tang, Yizeng Han, Zhenglin Wan, Fanqing Meng, Wei Wang, Bohan Zhuang, Wangbo Zhao, Yang You. *Agent-as-a-Router: Agentic Model Routing for Coding Tasks*. Under review.
- **Pengfei Zhou**, Zhiwei Tang, Xiaopeng Peng, Chenrui Zhou, Lama Moukheiber, Yixing Ma, Jiajun Song, Zhenglin Wan, Wangbo Zhao, Jiasheng Tang, Bohan Zhuang, Fan Wang, and Yang You. *Improving Generalization Robustness of Multimodal RLVR*. Under review.
- **Pengfei Zhou**, Jiajun Song, Zhiwei Tang, Yixing Ma, Xiaopeng Peng, Donghui Si, Yuhang Xu, Huiqi Song, Yiyuan Miao, Yichen Qian, Weihua Chen, Wangbo Zhao, Bohan Zhuang, Jiasheng Tang, Yang You. *Unified Hallucination Fuzzing for Multimodal Large Language Models*. Under review.
- Enqiao Lu, Zhenglin Wan, Xingrui Yu, **Pengfei Zhou**, Wangbo Zhao, Yang You, and Ivor Tsang. *Verifiable World Action Models: Logic-Graph Reasoning for Long-Horizon Robotic Manipulation*. Under review.

Patents

- **Pengfei Zhou**, and Cong Bai. A Cross-Modal Image Retrieval Method for Lifelogging, CN110750663B, 2021
- **Pengfei Zhou**, and Cong Bai. A Real-Time Diet Health Monitoring Method for Diabetic Patients based on Multimodal Data, CN111599438B, 2023
- Cong Bai, and **Pengfei Zhou**. A Cross-Modal Retrieval Method based on Graph Convolutional Neural Network, CN111598214B, 2023
- Weiqing Min, **Pengfei Zhou**, Ying Jin, and Shuqiang Jiang. Multi-task intelligent dietary management method, device, storage medium, computer equipment, CN118486430A, 2024
- Shuqiang Jiang, Yifan Yi, Weiqing Min, **Pengfei Zhou**, and Tianyu Zhang. A training method for the incremental food detection model. CN120107751A, 2025
- Weiqing Min, Chunlin Liu, **Pengfei Zhou**, Tao Liu, and Shuqiang Jiang. The Invention Discloses A Food Detection System, A Model Training Method and A Food Detection Method, 2022116988701, 2022

WORK EXPERIENCE

Research Intern

Alibaba DAMO Academy, Hangzhou, China

Aug. 2025 – Jun. 2026

- Conducted research on hallucination and post-training robustness of MLLMs.
- Developed new benchmark frameworks for assessing hallucination and robustness under fuzzing scenarios. Designed automated evaluation pipelines and large-scale datasets for stress-testing multimodal reasoning abilities. Analyzed failure modes and improved the generalization robustness of RLVR algorithms.
- Two related papers were finished and submitted to ICML 2026.

Research Intern

Shanghai Artificial Intelligence Laboratory, Shanghai, China

Jul. 2024 – Jul. 2025

- Led research on multimodal large language models (MLLMs) for interleaved image-text generation and evaluation.
- Proposed *OpenING*, the first comprehensive benchmark for open-ended interleaved image-text generation. The work was accepted as an **Oral at CVPR 2025 (0.7% of 13K submissions)**.
- Studied multimodal knowledge-point retrieval-augmented generation and chain-of-thought strategies to benchmark reasoning and task-solving capabilities of MLLMs. The work was published at AAAI 2026.
- Explored neural-driven generation by integrating EEG signals with diffusion-based image editing. Developed a multimodal processing and fusion algorithm for robust generation. The work was published at NeurIPS 2025.

Technical Director

Institute of Intelligent Computing Technology, CAS, Suzhou, China

Apr. 2022 – Oct. 2023

- Led a team of 8 employees to develop a visual system for dietary management and nutritional estimation
- Proposed food detection algorithm via zero-shot learning techniques with 99%+ accuracy in real-world scenarios
- Deployed dietary systems in Changzhou First People's Hospital, Puxiang Co. LTD and Xujiang Experimental Middle School with 100M+ usage, leading to a 3× checkout rate and 80% manpower requirement reduction
- Authorized a Chinese patent, published related papers in ACM MM 2023 and IEEE TIP

Co-founder, CTO

Hangzhou Xiao Bin Technology Co. LTD, Hangzhou, China

Jan. 2020 – May 2021

- Director of Smart Home System Research and Development Laboratory with 10+ employees
- Developed a smart bin recognition system (*99%+ classification accuracy, 30%+ higher than competitors*)
- Sold products in major cities in China including Shanghai and Hangzhou, with an annual turnover of CNY 100K+
- Won 10+ prizes in competitions including Silver Award in Zhejiang International "Internet+" Undergraduate Entrepreneurship Competition (*2/120*) and First Prize in National Undergraduate IoT Design Contest (*5/100*)

ACADEMIC ACTIVITIES

Reviewer: IEEE TNNLS/TMM/TCYB, ACM ToMM, NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, AAAI, ACM MM

Co-organizer: China Multimedia 2023 Conference (*1K+ attendees*): Website designer and system manager

RESEARCH INTERESTS

Current: Working on Multimodal Intelligence (*MLLMs*) and Multimedia Analysis (*Real-world Applications*)

Interested in: Generative Models (*LLMs, Diffusion*), AI-enabled Healthcare, and Open-world Topics (*World Models*)